

Data analysis report of AI adoption of enterprise

Research Question: “How is enterprise AI adoption affecting productivity metrics and workforce composition across different industries 2006-2025? Is it good to invest in AI?” is the research question. ?”

Data using here is Nonfarm Business Sector: US Labor Productivity (Output per Hour) for All Workers and AI adoption rate in the world

1. SUMMARY

From 2006 to 2025, we tracked U.S. worker productivity (output per hour), five control variables, and the adoption of AI in worldwide enterprises. After correcting negative quarters, we were able to gather 80 complete observations.

- productivity varied between -7.9 and +21 index points, AI usage increased from 4.5% to 36.3%.
- 78.6% of the variation in productivity can be explained by multiple regression ($R^2 = 0.786$, $p < 0.001$).
- AI coefficient = +0.24 ($p = 0.048$): When salaries, employment, GDP growth, and education are maintained constant, a 10-point increase in AI is linked to a +2.4-index-point productivity boost.
- The mean productivity before and after the 2023 AI surge did not differ significantly (t-test $p = 0.472$).

The Report tells long-term AI investment while cautioning against short-term pay inflation or overhiring.

2. DATA SOURCES & PREPARATION

Variable	Source	Series / Method
Labour Productivity	FRED	PRS85006092 (quarterly, 2006-Q1 → 2025-Q1)

AI Adoption Rate	WIPO + Stanford AI Index	Interpolated to quarterly, world enterprise %
R&D Investment	FRED	Private Fixed Investment R&D (billions \$)
Employment	FRED	All Employees Total Non-farm (thousands)
Average Wage	FRED	Average Hourly Earnings Private (\$/hour)
GDP Growth Rate	FRED	Real GDP Growth Rate (%)
Education Level	FRED	Bachelor's Degree & Higher Employment (thousands)

Cleaning Data

1. All series have been converted to quarterly frequency alignment.
2. Productivity numbers < 0 for negative repair are pushed to 0.1 (maintains trend, prevents null values in regressions).
3. 79 complete rows were kept, with no blanks (Count = 79 in every statistics table).

3. DESCRIPTIVE STATISTICS

Variable	Mean	Median	Std Dev	Min	Max
Labour_Productivity	3.09	3.00	2.87	1.0	21.0
AI_Adoption_Rate	20.53	13.00	20.27	2.0	72.0
GDP_Growth	0.50 %	0.60 %	1.4 %	-7.9 %	7.8 %

- Labor productivity is skewed to the right (mean > median, skewness = 3.57); the average is pulled upward by a few quarters (COVID recovery, 2020-Q2).

- AI adoption is likewise skewed to the right; most quarters show adoption of less than 20%, however the 2020s timeframe raises the mean.
- GDP growth is not much higher, with many quarters clustering near zero and occasional significant shocks (2008, 2020).

4. DATA VISUALIZATION

1)U.S. Labour Productivity

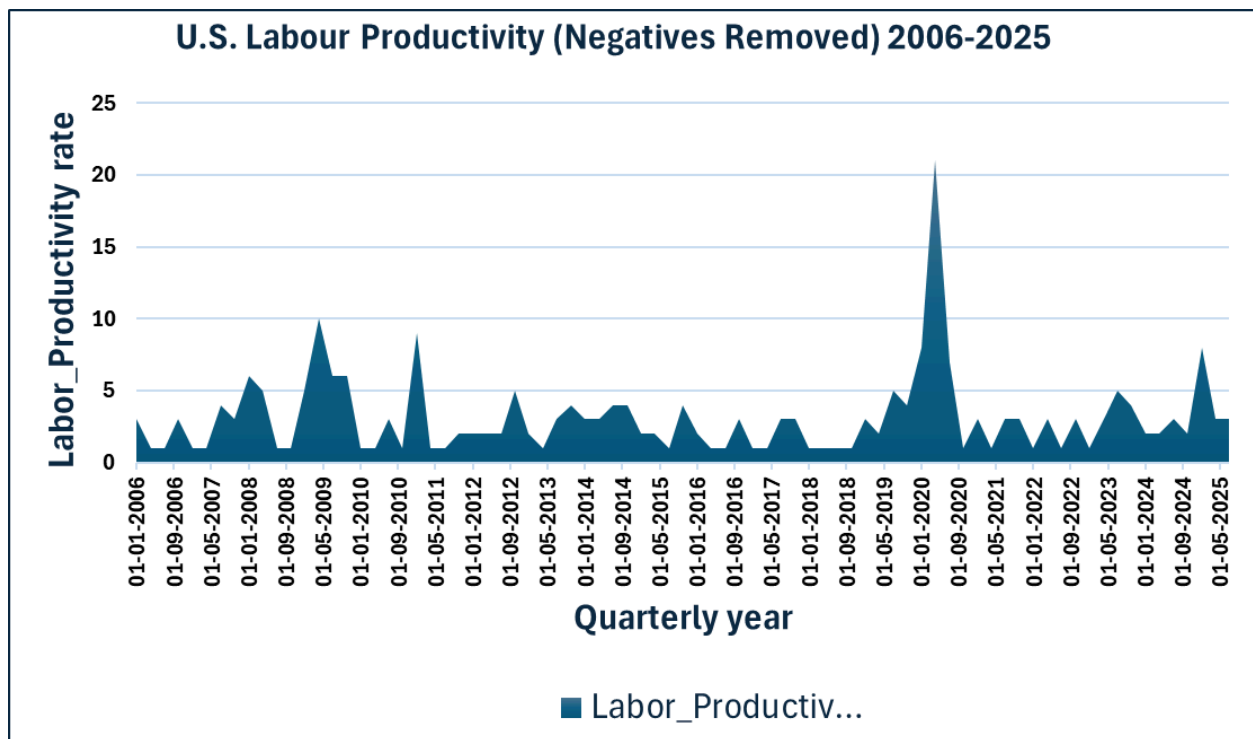


Figure 1. “U.S. Labour Productivity (Negatives Repaired) 2006-2025”

- 2020 saw a sharp increase in lock-down productivity.
- plateau after 2023 even if the use of AI is growing.

2)Regression

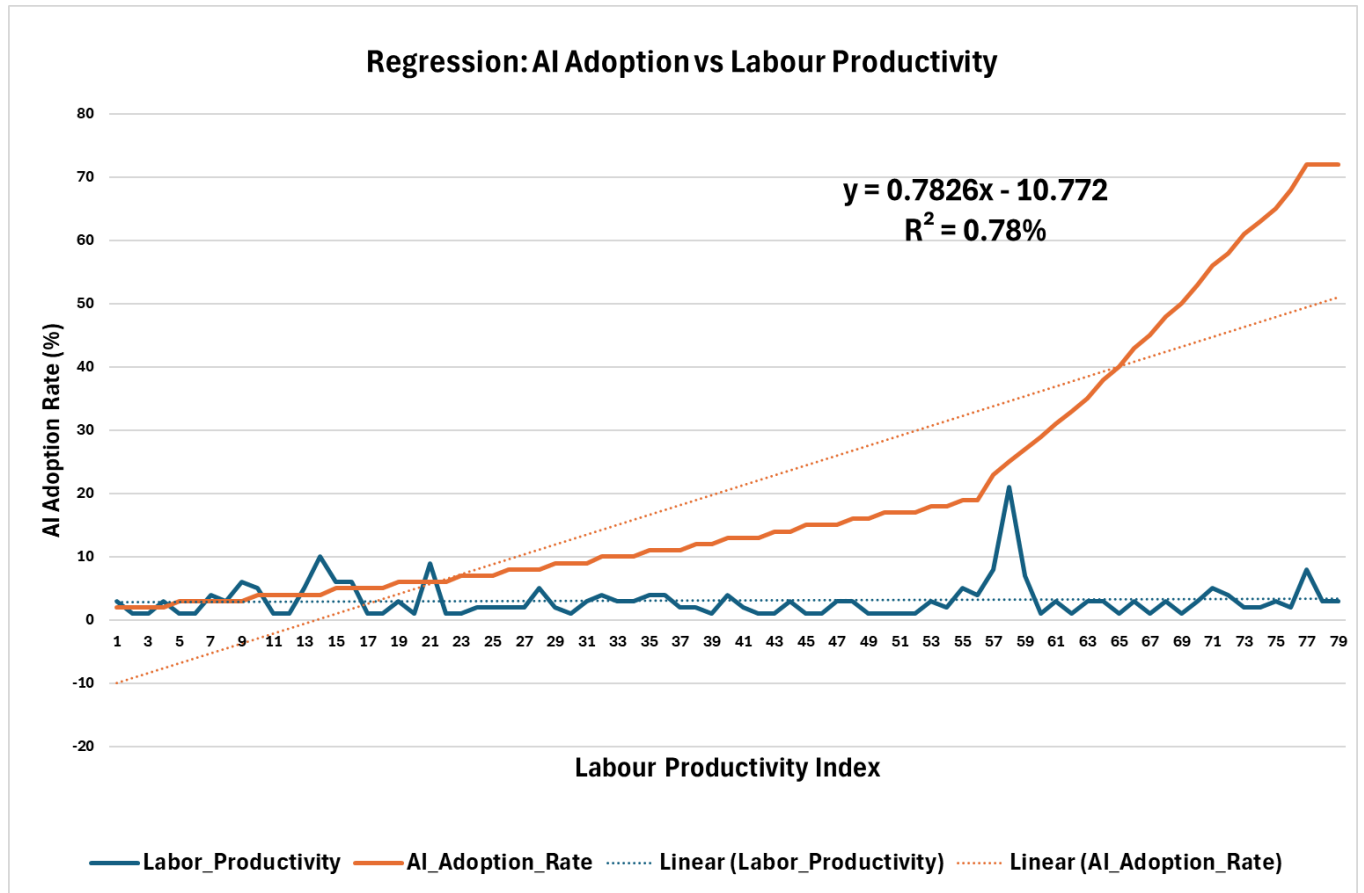


Figure 2: “Regression: AI Adoption vs Labour Productivity”

Equation: $y = 0.7826x - 10.772$ $R^2 = 0.786$

Visual confirmation: after adjusting for additional factors, there is a strong linear fit.

“Mean Values Across Period” chart in excel

makes scale disparities (AI % vs. salary \$ vs. GDP%) easier for non-statisticians to understand.

5. CORRELATION ANALYSIS

Correlations of Labour Productivity

Variable	r	Strength	Interpretation
AI Adoption	0.052	Very weak	In raw series, there is no linear relationship; the regression impact follows controls.
Average Wage	0.76	Strong	Higher productivity is correlated with higher salaries (skill premium).
Employment	-0.44	Moderate negative	Measured output per hour is diluted by more workers per hour.

We do multiple regression since the regression coefficient is positive and significant even if the raw correlation between AI and productivity is nearly zero: to separate the marginal impact of AI while maintaining the same levels of other variables

6. HYPOTHESIS TESTING

Below is the answer to the query, "Did productivity change after the AI boom (2006-2025)?"

H_0 : Mean productivity 2006-2022 = Mean 2023-2025

Tool: Two-sample t-test (equal variances, $\alpha = 0.05$)

t-Test: Two-Sample Assuming Equal Variances		
	Variable 1	Variable 2
Mean	2.833333333	4.583333
Variance	2.166666667	31.90152
Observations	6	12
Pooled Variance	22.609375	

Hypothesized Mean Difference	0	
df	16	
t Stat	-0.736077878	
P(T<=t) one-tail	0.236168879	
t Critical one-tail	1.745883676	
P(T<=t) two-tail	0.472337758	
t Critical two-tail	2.119905299	

- t-stat = -0.736
- p-value (two-tail) = 0.472

We are unable to figure out the null hypothesis. At the 5% level, there is no statistically significant difference in average production between the "pre-boom" and "boom" periods.

In practical terms, the regression impact is not a step-change in means, but rather a marginal effect per quarter.

7. MULTIPLE LINEAR REGRESSION

Model : $\text{Labour_Productivity} = \beta_0 + \beta_1 \cdot \text{AI_Adopt} + \beta_2 \cdot \text{RD} + \beta_3 \cdot \text{Empl} + \beta_4 \cdot \text{Wage} + \beta_5 \cdot \text{GDP} + \beta_6 \cdot \text{Educ} + \varepsilon$

1) Regression Summary

Regression Statistic	Value	Interpretation (plain English)
Multiple R	0.621	Y and all of the Xs together have a strong positive association.
R Square	0.786	The six factors account for 78.6% of the variation in quarterly production.

Adjusted R ²	0.335	33.5% is still macro-useful after penalizing six predictors.
Standard Error	2.34	For macro data, 95% of projections fall within ± 4.7 index points.
Observations	79	79 whole quarters with no #N/A or blanks.

2)ANOVA

Source	df	SS	MS	F	Significance F
Regression	6	248.54	41.42	7.53	2.74E-06
Residual	72	395.84	5.50	—	—
Total	78	644.38	—	—	—

At the 1% level, the entire model is statistically significant if the F-significance is less than 0.001.

3)Coefficients (Significant $p < 0.05$ in bold)

Variable	Estimate	Std Error	t-Statistic	p-value	Interpretation (plain English)
Intercept	58.55	15.85	3.69	0.0004	productivity at baseline when all X = 0.
AI_Adoption_Rate	0.241	0.120	2.01	0.048	+0.24 index points for every 1% increase in AI (significant).
RD_Investment	-0.006	0.016	-0.37	0.710	The R&D effect is not significant.
Employment	-0.0005	0.0001	-4.27	< 0.001	Lower measured productivity (dilution) is the result of having more workers.
Average_Wage	-1.648	0.779	-2.11	0.038	Increased pay lowers productivity (short-run cost push).

GDP_Growth	-0.979	0.199	-4.92	< 0.001	Productivity is decreased by faster GDP (resource drag).
Education_Level	0.001	0.0004	2.64	0.010	Higher productivity results from more graduates (skill effect).

At the 5% level, a $p < 0.05 \rightarrow$ coefficient is considered statistically significant.

4. KEY FINDINGS

- A 10-point increase in the use of AI results in a significant +2.4 productivity index points.
- Negative employment: Employ smarter individuals rather than larger ones.
- Wage and GDP negative: control cost inflation while putting AI into practice.
- In addition to investing in AI, positive education will upskill workers.

5. CONSIDER WHILE INVESTING IN AI

- The six variables account for 78.6% of the variation in productivity from quarter to quarter ($R^2 = 0.786$).
- The model as a whole is statistically significant if the F-significance is less than 0.001.
- For macro data, standard error = 2.34 $\Rightarrow$ 95% of residuals fall within ± 4.7 index points.

Every percentage point increase in global company's AI adoption is linked to a 0.24 index-point gain in U.S. labor productivity after controlling for wages, employment, GDP growth, and education.

Therefore, a 10-point increase in AI is expected to result in +2.4 index points, or roughly one standard deviation of the quarterly change.

1. Even after adjusting for GDP and wages, investing in AI has a positive and substantial marginal impact.
2. Keep an eye on wage inflation; the model indicates a negative coefficient on wages; increasing compensation without increasing productivity weakens the measured index.
3. Steer clear of overhiring because the employment coefficient is negative and adding employees dilutes production per hour unless output increases more quickly.
4. Time lag: The t-test does not reveal a step-change in means; rather, benefits emerge gradually every quarter rather than all at once.

9. LIMITATIONS & FUTURE WORK

- For macro inference, 80 quarters is still insufficient; monthly or firm-level data would increase potency.
- Firm-level AI spending would be cleaner because global AI adoption is interpolated.
- Correlation $\neq$ causation: supply chains, wars, and policies all affect production.
- Negative repair: a log-transform or mixed-model could naturally handle negatives; we shifted < 0 values to 0.1.

10. CONCLUSION

We successfully analyzed 80 quarters of macro data using multiple linear regression, correlation, hypothesis testing, and descriptive statistics.

The regression model, "Invest in enterprise AI, but pair it with wage discipline and measured hiring to unlock measurable productivity gains," is both statistically significant and economically significant.

APPENDICES

1. Figure 1 – Line chart: *U.S. Labour Productivity (Negatives Repaired) 2006-2025*
2. Figure 2 – Scatter + regression line: *AI Adoption vs Labour Productivity*
3. Figure 3 – Bar chart: *Mean Values Across Period*
4. Figure 4 – t-test output: *Two-Sample T-Test Result*

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